Curriculum Vitae

Santiago Coelho

October 29, 2026

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Education

Ph.D. in Biomedical Engineering 2015–2019
University of Leeds (2018–2019) and University of Sheffield (2015–2018), UK. Advisor: Prof. Alejandro Frangi.

Bachelor in Electrical Engineering 2010–2015
Five-year program, Universidad Nacional de La Plata (UNLP), La Plata, Argentina. Engineering Faculty Salutatorian.

Research and Teaching Experience

Assistant Professor (Research) Aug. 2026–present
Center for Biomedical Imaging (CBI), Department of Radiology, School of Medicine, New York University.

Postdoctoral Research Fellow Jan. 2020–July 2026
MRI Biophysics Group (PIs: Dmitry Novikov and Els Fieremans), Center for Biomedical Imaging (CBI), Department of Radiology, School of Medicine, New York University.

Teaching Instructor 2021–present
Practical Magnetic Resonance Imaging II (Ph.D. course), Department of Radiology, School of Medicine, New York University.

Graduate Teaching Assistant 2017–2018
Mathematics of Imaging Sciences (M.Sc. course), Department of Electronic and Electrical Engineering, University of Sheffield.

Teaching Assistant 2012–2015
Numerical Analysis, Faculty of Engineering, Universidad Nacional de La Plata (FiUNLP).

Teaching Assistant 2014
Control and Servomechanisms, FiUNLP.

Teaching Assistant 2013
Electromagnetism, FiUNLP.

Teaching Assistant 2011
Calculus II, FiUNLP.

Honors and Awards

- **Junior Fellow of the ISMRM** (2025).
- **NIH K99/R00 Award** (K99EB036080-01), NIBIB, NIH; approximately $1M for 2024–2029.
- Magna Cum Laude Award, ISMRM Annual Meeting, work #0641 (2024).
- Summa Cum Laude Award and AMPC Top 100 selection, ISMRM Annual Meeting, work #0069 (2023).
- Best Presentation Award, ISMRM Diffusion Study Group Workshop (2022).
- Magna Cum Laude Award, ISMRM Annual Meeting, work #0514 (2022).
- Best Presentation Award, ISMRM Diffusion Study Group, ISMRM Annual Meeting (2022).
- Best Presentation Award, ISMRM White Matter Study Group, ISMRM Annual Meeting (2022).
- Summa Cum Laude Award, ISMRM Annual Meeting, work #0397 (2021).
- Best Presentation Award, ISMRM Diffusion Day Trainee Workshop (2021).
- Award to the Best Graduate of Electrical Engineering, UNLP, La Plata, Argentina (2015).
- Annual Award from the Professional Council of Mechanical and Electrical Engineering for the highest GPA (2015).
- J. V. Gonzalez Award, La Plata City Council, for achieving one of the highest GPAs in FiUNLP (2015).
- Best Graduate Award, FiUNLP, for achieving the highest GPA in an engineering specialty (2015).
- Luis A. Huergo Award, Buenos Aires College of Engineers, for the highest GPA in a UNLP engineering specialty (2015).

Leadership

- **Elected Trainee Representative**, ISMRM Diffusion Study Group (approximately 1,000 members), 2023–2025.
- **Chair**, Tissue Microstructure Imaging Gordon Research Seminar (GRS), 2023.
  Held in conjunction with the namesake Gordon Research Conference, the seminar provided a forum for doctoral and postdoctoral researchers to present work, discuss emerging methods and pre-publication data, and build collaborations. It covered multimodal approaches to tissue imaging at the cellular level, including microscopy, ultrasound, and MRI.
- **Co-Chair**, ISMRM Diffusion Trainee Day, 2024.
  This 24-hour virtual workshop spanned three time zones and included educational lectures, scientific and poster presentations, and networking activities for early-career researchers.
- **Co-supervisor**, M.Sc. project, "Multiple approaches for Diffusion Tensor Imaging estimation" (2017). Main advisor: Prof. Alejandro Frangi.

Service and Editorial Activities

- **Organizing Committee**, ISMRM Diffusion Study Group Workshop, "40 Years of Diffusion: Past, Present & Future Perspectives", Kyoto, February 2025. Co-Chair of the Educational and Historical Perspective Programs.
- **Organizing Committee**, ISMRM Member-Initiated Session, "Pushing the Boundaries of Diffusion MRI Across Field Strengths & Gradient Performance" (MIS-10), Singapore, 2024.
- **Guest Editor**, virtual special issue of *Magnetic Resonance in Medicine* on Diffusion MR (2023).
- **Co-Managing Guest Editor**, special issue of *Imaging Neuroscience* on Diffusion MRI (2025–2026).
- **Reviewer** for *NeuroImage*, *Magnetic Resonance in Medicine*, *Imaging Neuroscience*, *Communications Biology*, *Human Brain Mapping*, *Medical Image Analysis*, *NMR in Biomedicine*, *Scientific Reports*, and *American Journal of Neuroradiology*.

Patents

1. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. "System, method and computer-accessible medium for determining rotational invariants of cumulant expansion from one or more acquisitions which can be minimal." U.S. Provisional Application 63/333,856; converted to a full PCT application April 25, 2023.
2. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. "System, method and computer-accessible medium for diffusion MRI without shells." Provisional Patent Application 63/412,181; converted to a full PCT application September 30, 2023.

Peer-Reviewed Publications

1. **Santiago Coelho**, Jenny Chen, Filip Szczepankiewicz, Els Fieremans, and Dmitry S. Novikov. "Geometry of the cumulant series in diffusion MRI." *Nature Communications*, 17:4220, 2026.
2. **Santiago Coelho**, . . . , Thomas K. F. Foo, Els Fieremans, and Dmitry S. Novikov. "What if each voxel were measured with a different diffusion protocol?" *Magnetic Resonance in Medicine*, 2025.
3. Sohae Chung, . . . , **Santiago Coelho**, . . . , and Yvonne W. Lui. "Linking Symptom Phenotypes to Patterns of White Matter Injury in Mild Traumatic Brain Injury: A Latent Class Analysis." *American Journal of Neuroradiology*, 2025.
4. Kwok-Shing Chan, Yixin Ma, Hansol Lee, José P. Marques, Jonas Olesen, **Santiago Coelho**, et al. "In vivo human neurite exchange time imaging at 500 mT/m diffusion gradients." *Imaging Neuroscience*, 3:1–26, 2025.
5. Benjamin Ades-Aron, **Santiago Coelho**, et al. "Denoising Improves Cross-Scanner and Cross-Protocol Test–Retest Reproducibility of Diffusion Tensor and Kurtosis Imaging." *Human Brain Mapping*, 46:e70142, 2025.
6. **Santiago Coelho**, Ying Liao, et al. "Assessment of precision and accuracy of brain white matter microstructure using combined diffusion MRI and relaxometry." *Human Brain Mapping*, 2024.
7. Ricardo Coronado-Leija, Ali Abdollahzadeh, Hong-Hsi Lee, **Santiago Coelho**, et al. "Volume electron microscopy in injured rat brain validates white matter microstructure metrics from diffusion MRI." *Imaging Neuroscience*, 2024.
8. Ying Liao, **Santiago Coelho**, Jenny Chen, et al. "Mapping tissue microstructure of brain white matter in vivo in health and disease using diffusion MRI." *Imaging Neuroscience*, 2024.
9. Sohae Chung, Tamar Bacon, Joseph F. Rath, Alaleh Alivar, **Santiago Coelho**, et al. "Callosal Interhemispheric Communication in Mild Traumatic Brain Injury: A Mediation Analysis on WM Microstructure Effects." *American Journal of Neuroradiology*, 2024.
10. **Santiago Coelho**, Steven H. Baete, Gregory Lemberskiy, et al. "Reproducibility of the standard model of diffusion in white matter on clinical MRI systems." *NeuroImage*, 2022.
11. L. A. Wilson, L. Heraty, B. A. Ashford, **Santiago Coelho**, A. F. Frangi, J. M. Pozo, P. G. Ince, and J. R. Highley. "Tissue

microarray (TMA) use in post mortem neuropathology." *Journal of Neuroscience Methods*, 2021.

12. Meghdoot Mozumder, Jose M. Pozo, **Santiago Coelho**, and Alejandro F. Frangi. "Population-based Bayesian regularization for microstructural diffusion MRI with NODDIDA." *Magnetic Resonance in Medicine*, 2019.
13. **Santiago Coelho**, Jose M. Pozo, et al. "Histological data of axons, astrocytes, and myelin in deep subcortical white matter populations." *Data in Brief*, 2019.
14. **Santiago Coelho**, Jose M. Pozo, et al. "Resolving degeneracy in diffusion MRI biophysical model parameter estimation using double diffusion encoding." *Magnetic Resonance in Medicine*, 2019.
15. Rachel Waller, Lynne Baxter, Daniel J. Fillingham, **Santiago Coelho**, et al. "Iba-1-/CD68+ microglia are a prominent feature of age-associated deep subcortical white matter lesions." *PLoS ONE*, 2019.
16. Meghdoot Mozumder, Jose M. Pozo, **Santiago Coelho**, et al. "Quantitative histomorphometry of capillary microstructure in deep white matter." *NeuroImage: Clinical*, 2019.
17. **Santiago Coelho**, Jose M. Pozo, et al. "Local volume fraction distributions of axons, astrocytes, and myelin in deep subcortical white matter." *NeuroImage*, 2018.

Book Chapters

1. **Santiago Coelho**, Yogesh Rathi, and Marco Reisert. "Model Estimation," in *Probing Brain Microstructure with MRI*, Springer Nature's Neuroscience Methods series. Edited by Sune Jespersen and Noam Shemesh; to be published in 2027. *In preparation.*
2. Els Fieremans, Kouhei Kamiya, and **Santiago Coelho**. "Standard Diffusion MRI," in *Diffusion MRI User Guide: A Window to Biological Tissue*, Springer Nature's Diagnostic Imaging Series. Edited by Mami Iima, Susie Huang, and Denis Le Bihan; to be published in 2026. *Accepted; in editorial production.*

Preprints and Publications Under Review

1. Valentin Stepanov, **Santiago Coelho**, Benjamin Ades-Aron, Jenny Chen, Timothy Shepherd, Ilya Kister, Dmitry Novikov, and Els Fieremans. "Advanced quantitative diffusion MRI markers unveil changes in the normal-appearing white matter reflecting disability in multiple sclerosis." *In preparation; planned for submission in 2026.*
2. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. "Zero-Shell Imaging (ZSI): microstructure-informed undersampling of diffusion MRI." *In preparation; planned for submission in 2026.*
3. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. "How do we know we measure tissue parameter, not the prior?" *In preparation; planned for submission in 2026.*

Presentations at Scientific Conferences

Peer-Reviewed Conference Abstracts

Delivered 40 presentations at ISMRM meetings (2017–2025), including the following first- or senior-author oral presentations and power pitches:

1. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. "When fiber bundles are not axially symmetric: 4-dimensional fiber distributions in the human brain" Annual Meeting of the ISMRM, 2026.
2. Valentina Mazzoli, Smita Rao, . . . , Dmitry S. Novikov, Els Fieremans, and **Santiago Coelho**. "Microstructure and muscle strength in older adults: a time-dependent diffusion study." Annual Meeting of the ISMRM, 2025.
3. **Santiago Coelho**, Filip Szczepankiewicz, Els Fieremans, and Dmitry S. Novikov. "Cumulant Tensors From The Addition Of Angular Momenta: All Diffusion Invariants In One Abstract." Annual Meeting of the ISMRM, 2024.
4. **Santiago Coelho**, Valentin Stepanov, et al. "Sticks Or No Sticks? White Matter Microstructure In Multiple Sclerosis From High-B Scaling." Annual Meeting of the ISMRM, 2024.
5. **Santiago Coelho**, Gregory Lemberskiy, et al. "What If Every Voxel Was Measured With A Different Diffusion Protocol?" Annual Meeting of the ISMRM, 2023.
6. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. "Zero-Shell Diffusion MRI: Focus On Microstructure By Decoupling Fiber Orientations." Annual Meeting of the ISMRM, 2023.
7. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. "Rotational Invariants Of The Cumulant Expansion (RICE)." Annual Meeting of the ISMRM, 2022.
8. **Santiago Coelho**, Filip Szczepankiewicz, Els Fieremans, and Dmitry S. Novikov. "Fast Acquisitions For Mean Kurtosis And Microscopic Fractional Anisotropy." Annual Meeting of the ISMRM, 2022.
9. **Santiago Coelho**, Filip Szczepankiewicz, et al. "Reproducibility Of White Matter Microstructure Mapping With Diffusion-

Relaxometry." Annual Meeting of the ISMRM, 2022.

10. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. "How do we know we measure tissue parameters, not the prior?" Annual Meeting of the ISMRM, 2021.
11. **Santiago Coelho**, et al. "Optimal experimental design in multidimensional diffusion MRI for parameter estimation of biophysical tissue models." Annual Meeting of the ISMRM, 2020.

Peer-Reviewed Conference Articles

1. **Santiago Coelho**, Jose M. Pozo, Sune N. Jespersen, and Alejandro F. Frangi. "Optimal experimental design for biophysical modelling in multidimensional diffusion MRI." *MICCAI Proceedings, Lecture Notes in Computer Science*, 2019.

Invited Talks

- **i2i NYU Workshop** — *Invited talk*, New York, October 2026.
- **White Matter Study Group Workshop** — *Educational lecture*, Marseille, France, October 2026.
- **BrainMap Seminars**, Martinos Center, Massachusetts General Hospital and Harvard Medical School — Boston, October 2025.
- **Department of Radiology, Weill Cornell Medicine** — New York, October 2025.
- **What Can You Be With a PhD? Symposium** — *Invited speaker*, New York, October 2025.
- **Institute of Imaging Science, Vanderbilt University** — Nashville, September 2025.
- **Department of Radiology, Columbia University** — New York, August 2025.
- **BioMedical Engineering and Imaging Institute, Icahn School of Medicine at Mount Sinai** — New York, April 2025.
- **Diffusion Microclub Scientific Seminars** — Virtual, October 2024.
- **ISMRM Annual Meeting** — *Educational lecture*, Toronto, Canada, June 2023.
- **ISMRM Diffusion Study Group Virtual Workshop** — *Educational lecture*, December 2021.
- **ISMRM Annual Meeting** — *Educational lecture*, virtual, May 2021.
- **Faculty of Engineering, Universidad Nacional de La Plata** — La Plata, Argentina, December 2019.